WIZARD

Robotic policy adaptation via weight-space meta-learning

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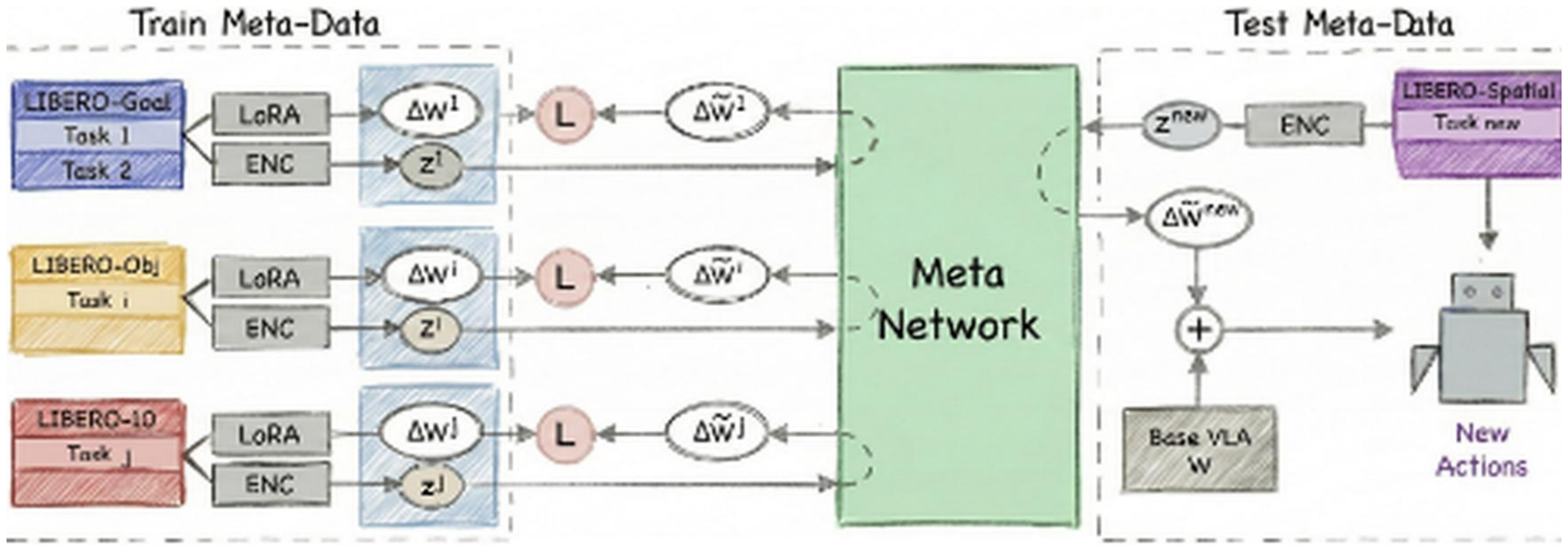
Results

LIBERO	π0.5 FT	LoRA-Gen	WIZARD
Spatial	19%	28%	**40%**
Object	1%	1%	**3%**
Goal	14%	17%	**22%**
10: part 1	3%	5%	**9%**
10: part 2	3%	4%	**7%**

MetaWorld ML45	Success rate
NN retrieval	**44%**
Fine-tuned π0.5	**58%**
LoRA-Gen	**60%**
WIZARD	**66%**

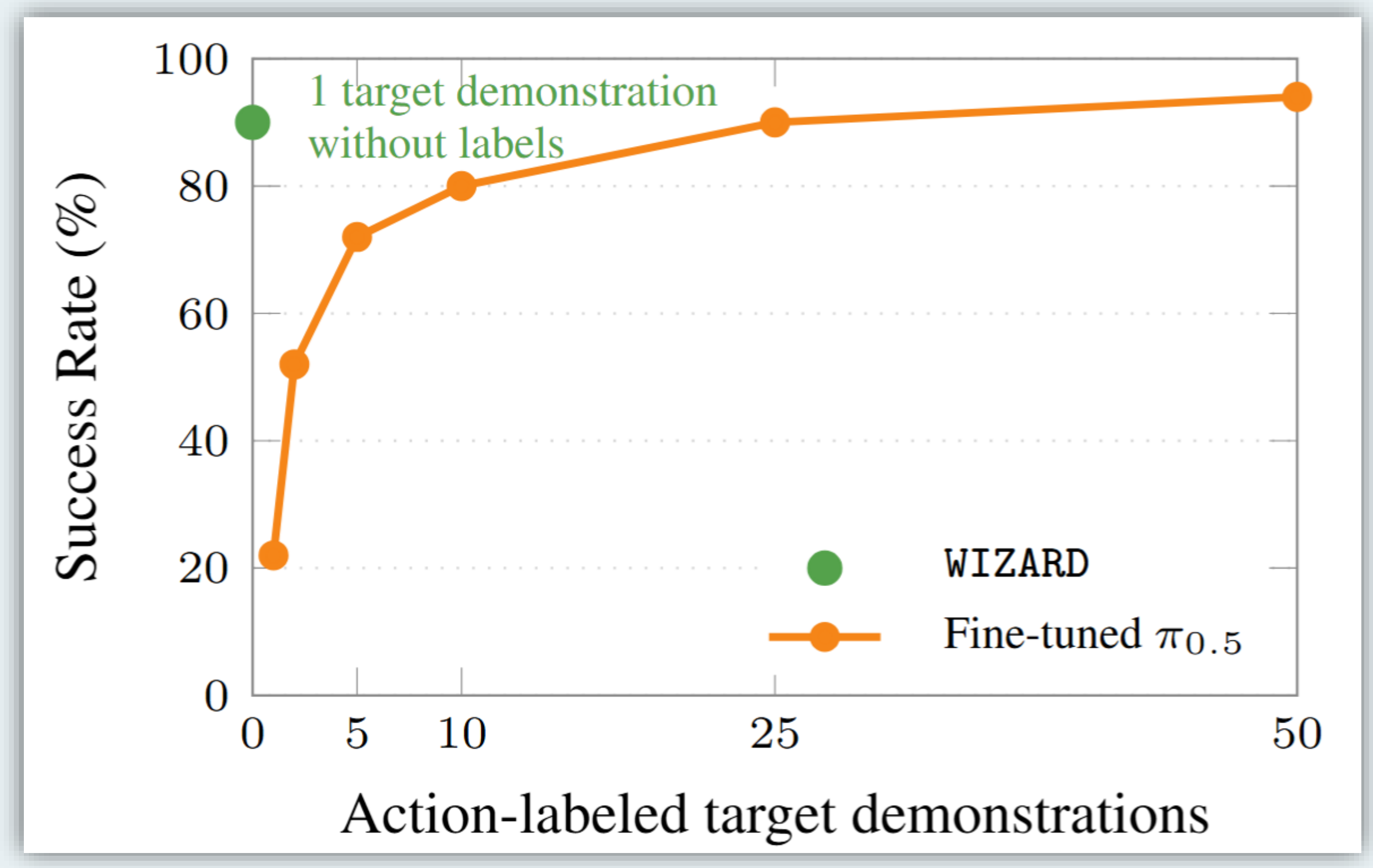
Real World	π0.5 FT	WIZARD
Success rate	17%	**37%**

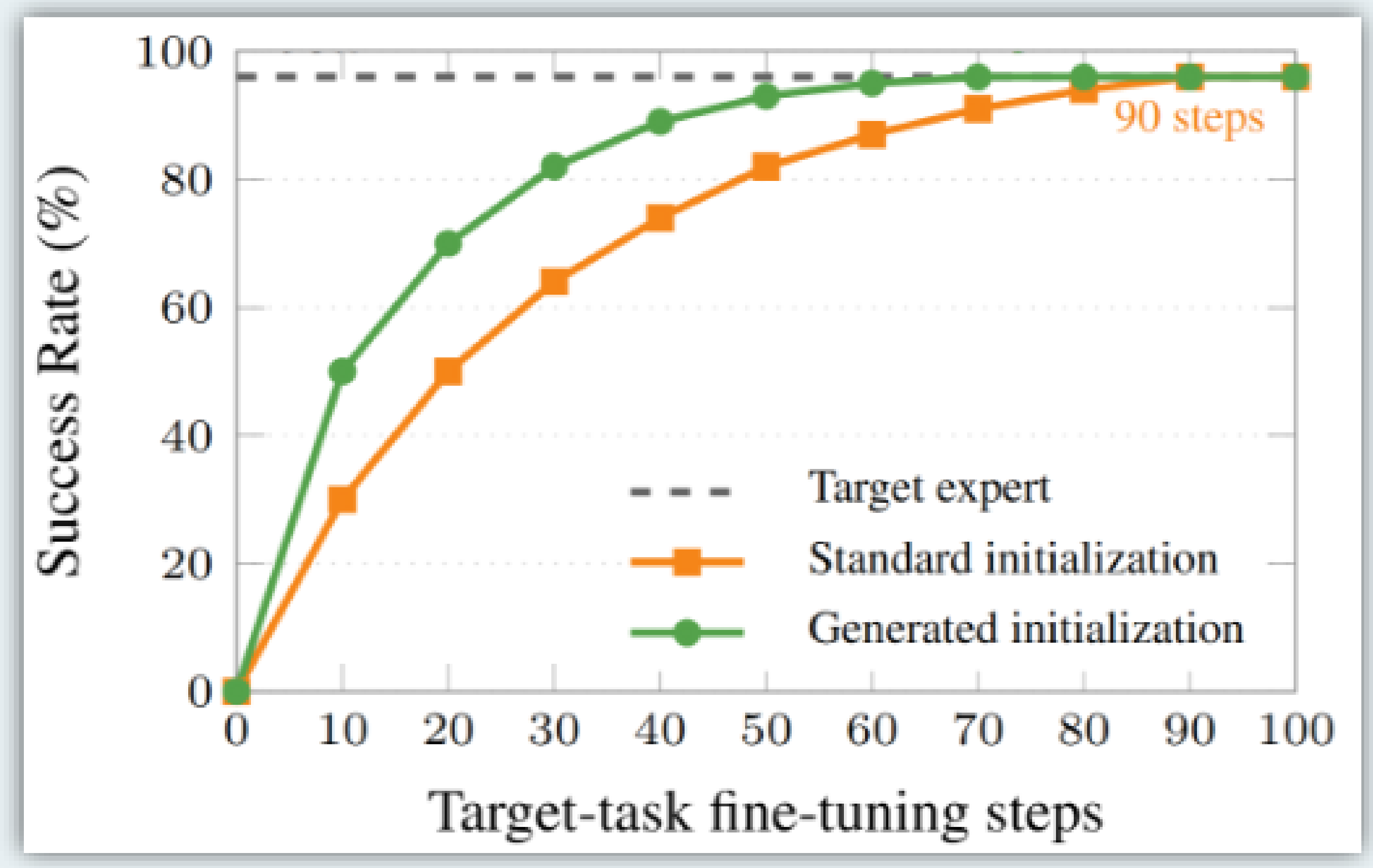


Overview

WIZARD generates LoRA updates for a frozen VLA using one visual demonstration and a language instruction. No target-task action labels or gradient updates. Source experts supervise adapter generation during training. At deployment, one instruction–video example generates a new adapter in a single forward pass.

Adaptation efficiency





References: π0.5 – Physical Intelligence, 2025 · LIBERO – Liu et al., NeurIPS 2023 · Meta-World – Yu et al., CoRL 2019
LoRA – Hu et al., ICLR 2022 · LoRA-Gen – Xiao et al., 2025

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